
ICPC Algorithm Library

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1 Basic

1.1 test

```
export fs="-fsanitize=address,undefined"
ulimit -s 1024000
ulimit -v 1024000
mk() { g++ -o $1 $1.cpp -O2; }
```

1.2 pai

```
pai() {
    mk a; mk bf; mk gen
    while true; do
        ./gen > 1.in
        ./bf < 1.in > 1.ans
        ./a < 1.in > 1.out
        if diff -Zq 1.out 1.ans; then echo ac; else echo wa; break; fi
    done
}
```

1.3 run

```
shopt -s nullglob # 没有匹配到任何文件时, 让它返回空
ulimit -s 1024000
g++ $1.cpp -o $1 -std=c++20 -O2 -fsanitize=address,undefined || exit
for f in "$2"/*.in; do
    x=${f%.in}
    \time -f "$x: %es %MKB" ./$1 < $x.in > $1.out
    diff -Zq $1.out $x.ans
done
```

1.4 缺省源

```
#include <bits/stdc++.h>
using namespace std;

#define ALL(s) s.begin(), s.end()
#define SZ(s) int(s.size())
#define pb push_back

using LL = long long;
using ULL = unsigned long long;
using I = __int128;

void Cas() {
}
int main() {
```

```
cin.tie(0)->sync_with_stdio(0);
int t = 1;
cin >> t;
while (t--) {
    Cas();
}
return 0;
}
```

1.5 随机数

```
ULL seed = chrono::steady_clock::now().time_since_epoch().count();
mt19937_64 rnd(seed);
```

1.6 Matrix

```
template <class T>
struct Mat {
    int n, m;
    vector<T> a;
    Mat(int n = 0, int m = 0, T v = {}) : n(n), m(m), a(n * m, v) {}
    T& operator()(int i, int j) { return a[i * m + j]; }
};

using M = Mat<LL>;
M mul(M a, M b) {
    M c(a.n, b.m);
    for (int i = 0; i < a.n; i++) {
        for (int j = 0; j < b.m; j++) {
            for (int k = 0; k < a.m; k++) {
                c(i, j) = max(c(i, j), a(i, k) + b(k, j));
            }
        }
    }
    return c;
}
```

2 Graph

2.1 Dijkstra

```
struct Node {
    int u;
    LL d;
    bool operator<(const Node& o) const {
        return d > o.d;
    }
};
```

```
priority_queue<Node> q;
vector<LL> D(n, INF);
D[0] = 0;
q.push({0, 0});
while (SZ(q)) {
    auto [u, d] = q.top();
    q.pop();
    if (d != D[u]) continue;
    for (auto [v, w] : G[u]) {
        if (D[v] > d + w) {
            D[v] = d + w;
            q.push({v, D[v]});
        }
    }
}
```

2.2 Tarjan SCC

```
int tim = 0, scc = 0;
vector<int> dfn(n), low(n), bel(n, -1), stk;
auto tarj = [&](auto _, int u) -> void {
    dfn[u] = low[u] = ++tim;
    stk.push_back(u);
    for (int v : G[u]) {
        if (!dfn[v]) {
            _(_, v);
            low[u] = min(low[u], low[v]);
        } else if (bel[v] == -1) {
            low[u] = min(low[u], dfn[v]);
        }
    }
    if (dfn[u] != low[u]) return;
    while (1) {
        int v = stk.back();
        stk.pop_back();
        bel[v] = scc;
        if (v == u) break;
    }
    scc++;
};
for (int u = 0; u < n; u++) {
    if (!dfn[u]) tarj(tarj, u);
}
```

2.3 2-SAT

- $u_i = 2 \times u + i$
- $u_a \vee v_b$: 加入 $u_{a \oplus 1} \rightarrow v_b$ 和 $v_{b \oplus 1} \rightarrow u_a$

- $u_a \Rightarrow v_b$: 加入 $u_a \rightarrow v_b$ 和 $v_{b \oplus 1} \rightarrow u_{a \oplus 1}$
- 强制 $u = a$: 加入 $u_{a \oplus 1} \rightarrow u_a$
- 禁止 $u = a$ 与 $v = b$ 同时成立: 加入 $u_a \rightarrow v_{b \oplus 1}$ 和 $v_b \rightarrow u_{a \oplus 1}$
- $u = v$: 加入 $u_0 \leftrightarrow v_0$ 和 $u_1 \leftrightarrow v_1$
- $u \neq v$: 加入 $u_0 \leftrightarrow v_1$ 和 $u_1 \leftrightarrow v_0$

其中每个 $\leftrightarrow$ 都表示正反两条有向边。

若存在 u 满足 u_0, u_1 在同一强连通分量中, 则无解。

构造: $\text{ans}_u = [\text{bel}_{u_0} > \text{bel}_{u_1}]$ 。

2.4 Segment Tree Graph

- 原图和区间均使用 0-index: 原图节点编号为 $0 \sim n-1$, $\text{mdf}(l, r, \dots)$ 中的 $[l, r]$ 为闭区间。
- ZKW 线段树内部仍使用从 1 开始的堆式编号, 叶子位置为 $S \sim 2S-1$, 原图节点 i 对应叶子 $S+i$ 。
- sn 的值为 $2S-1$, 表示一棵完整 ZKW 线段树的节点数。
- 对于 $1 \leq p \leq \text{sn}$, $\text{id}(p, 0)$ 为 $n+p-1$, 表示 down 树节点; $\text{id}(p, 1)$ 为 $n+\text{sn}+p-1$, 表示 up 树节点。
- tot 初始为 $n+2(2S-1)$, 表示当前节点数, 也是下一个可用节点编号; 初始有效编号为 $0 \sim \text{tot}-1$, $\text{Node}()$ 返回 tot 后将其加一。

```
int S = 1;
while (S < n) S <<= 1;
int sn = 2 * S - 1;
int tot = n + 2 * sn;
vector<vector<pair<int, LL>>> G(tot);
auto id = [&](int p, int up) {
    return n + p - 1 + up * sn;
};
auto add = [&](int u, int v, LL w) {
    G[u].push_back({v, w});
};
auto link = [&](int u, int v, LL w, int up) {
    if (up) swap(u, v);
    add(u, v, w);
};
auto Node = [&]() {
    G.emplace_back();
    return tot++;
};
for (int up = 0; up < 2; up++) {
    for (int p = 1; p < S; p++) {
        link(id(p, up), id(p * 2, up), 0, up);
        link(id(p, up), id(p * 2 + 1, up), 0, up);
    }
    for (int i = 0; i < n; i++) {
```

```

    link(id(S + i, up), i, 0, up);
}
}
auto mdf = [&](int l, int r, int x, LL w, int up) {
    for (l += S, r += S + 1; l < r; l >>= 1, r >>= 1) {
        if (l & 1) link(x, id(l++, up), w, up);
        if (r & 1) link(x, id(--r, up), w, up);
    }
};

```

```

// x -> [l, r]
mdf(l, r, x, w, 0);
// [l, r] -> x
mdf(l, r, x, w, 1);
// [l1, r1] -> [l2, r2]
int x = Node();
mdf(l1, r1, x, 0, 1);
mdf(l2, r2, x, w, 0);

```

3 Data Structure

3.1 单点修改线段树

- 0-index, 左闭右开 $[l, r)$ 。
- 本模板不将 n 补齐到 2 的幂, 不能直接读取 $t[1]$ 查询全局结果, 须使用 $\text{prod}(0, \hookrightarrow n)$ 。

```

template <class S> struct SGT {
    int n;
    vector<S> t;
    SGT(int n) : n(n), t(2 * n) {}
    SGT(vector<S> a) : SGT(SZ(a)) {
        copy(ALL(a), t.begin() + n);
        for (int p = n - 1; p; p--) t[p] = t[2 * p] + t[2 * p + 1];
    }
    void set(int p, S x) {
        t[p += n] = x;
        while (p >>= 1) t[p] = t[2 * p] + t[2 * p + 1];
    }
    S prod(int l, int r) {
        S x{}, y{};
        for (l += n, r += n; l < r; l >>= 1, r >>= 1) {
            if (l & 1) x = x + t[l++];
            if (r & 1) y = t[--r] + y;
        }
        return x + y;
    }
};

```

```

struct S {
    LL mx = -INF;
};
S operator+(S a, S b) {
    return {max(a.mx, b.mx)};
}

```

3.2 区间修改线段树

- 0-index, 左闭右开 $[l, r)$ 。
- 按题目补全 V 、 $L::\text{operator}==$ 、merge 和 apply, 其中 $L\{\}$ 为空标记。
- 多组测试每组先用 $\text{seg}=\{\}$; 清空整棵线段树, 再重新 build。

```

#define ls (p << 1)
#define rs (p << 1 | 1)
struct SGT {
    struct V {
    };
    struct L {
        bool operator==(L b) {}
    };
    int n;
    V t[N << 2];
    L lz[N << 2];
    V merge(V a, V b) {}
    void apply(int p, int l, int r, L x) {}
    void up(int p) {
        t[p] = merge(t[ls], t[rs]);
    }
    void down(int p, int l, int r) {
        if (lz[p] == L{}) return;
        int m = (l + r) / 2;
        apply(ls, l, m, lz[p]);
        apply(rs, m, r, lz[p]);
        lz[p] = {};
    }
    void build(const vector<V>& a, int p = 1, int l = 0, int r = -1) {
        if (r < 0) n = r = SZ(a);
        if (r - l == 1) {
            t[p] = a[l];
            return;
        }
        int m = (l + r) / 2;
        build(a, ls, l, m);
        build(a, rs, m, r);
        up(p);
    }
    void mdf(int ql, int qr, L x, int p = 1, int l = 0, int r = -1) {

```

```

    if (r < 0) r = n;
    if (ql <= l && r <= qr) return apply(p, l, r, x);
    down(p, l, r);
    int m = (l + r) / 2;
    if (ql < m) mdf(ql, qr, x, ls, l, m);
    if (m < qr) mdf(ql, qr, x, rs, m, r);
    up(p);
}
V qry(int ql, int qr, int p = 1, int l = 0, int r = -1) {
    if (r < 0) r = n;
    if (ql <= l && r <= qr) return t[p];
    down(p, l, r);
    int m = (l + r) / 2;
    if (qr <= m) return qry(ql, qr, ls, l, m);
    if (m <= ql) return qry(ql, qr, rs, m, r);
    return merge(qry(ql, qr, ls, l, m), qry(ql, qr, rs, m, r));
}
};
#undef ls
#undef rs

```

```

SGT seg;
seg = {};
seg.build(a);
seg.mdf(l, r, x);
auto ans = seg.qry(l, r);

```

3.3 线段树二分

- 将下列函数放进上一节 SGT; $g(V\{\})$ 为真, 且扩展区间后判定具有单调性。
- $\text{max_right}(l, g)$ / $\text{min_left}(r, g)$ 分别返回满足 $g(\text{qry}(l, r))$ 的最大 r / 最小 l 。

```

#define ls (p << 1)
#define rs (p << 1 | 1)
template <class G>
int max_right(int ql, G& g, V& x, int p, int l, int r) {
    if (r <= ql) return r;
    if (ql <= l) {
        V y = merge(x, t[p]);
        if (g(y)) {
            x = y;
            return r;
        }
    }
    if (r - l == 1) return l;
}
down(p, l, r);
int m = (l + r) / 2;
int q = max_right(ql, g, x, ls, l, m);
if (q < m) return q;

```

```

    return max_right(ql, g, x, rs, m, r);
}
template <class G> int max_right(int l, G g) {
    V x{};
    return max_right(l, g, x, 1, 0, n);
}
template <class G>
int min_left(int qr, G& g, V& x, int p, int l, int r) {
    if (qr <= l) return l;
    if (r <= qr) {
        V y = merge(t[p], x);
        if (g(y)) {
            x = y;
            return l;
        }
    }
    if (r - l == 1) return r;
}
down(p, l, r);
int m = (l + r) / 2;
int q = min_left(qr, g, x, rs, m, r);
if (q > m) return q;
return min_left(qr, g, x, ls, l, m);
}
template <class G> int min_left(int r, G g) {
    V x{};
    return min_left(r, g, x, 1, 0, n);
}
#undef ls
#undef rs

```

3.4 Chairman Tree

- 值域使用 $0 \sim m-1$; 依次调用 $\text{add}(x)$ 后, $\text{rt}[i]$ 表示前 i 个元素形成的版本。
- $\text{count}(l, r, ql, qr)$ 统计下标区间 $[l, r)$ 中值落在 $[ql, qr)$ 的元素个数。
- $\text{kth}(l, r, k)$ 查询 $[l, r)$ 的第 k 小值域下标, k 从 0 开始。

```

struct Chairman {
    struct Node {
        int ls, rs, sum;
    };
    int n;
    vector<Node> t = {{0, 0, 0}};
    vector<int> rt = {0};
    Chairman(int n) : n(n) { assert(n > 0); }
    int copy(int p) {
        t.push_back(t[p]);
        return SZ(t) - 1;
    }
    int add(int p, int l, int r, int x, int v) {

```

```

int q = copy(p);
t[q].sum += v;
if (r - l == 1) return q;
int m = (l + r) / 2;
if (x < m) t[q].ls = add(t[q].ls, l, m, x, v);
else t[q].rs = add(t[q].rs, m, r, x, v);
return q;
}

void add(int x, int v = 1) {
    assert(0 <= x && x < n);
    rt.push_back(add(rt.back(), 0, n, x, v));
}

int count(int a, int b, int l, int r, int ql, int qr) {
    if (qr <= l || r <= ql) return 0;
    if (ql <= l && r <= qr) return t[b].sum - t[a].sum;
    int m = (l + r) / 2;
    return count(t[a].ls, t[b].ls, l, m, ql, qr)
        + count(t[a].rs, t[b].rs, m, r, ql, qr);
}

int count(int l, int r, int ql, int qr) {
    assert(0 <= l && l <= r && r < SZ(rt));
    assert(0 <= ql && ql <= qr && qr <= n);
    return count(rt[l], rt[r], 0, n, ql, qr);
}

int kth(int a, int b, int l, int r, int k) {
    if (r - l == 1) return l;
    int m = (l + r) / 2;
    int s = t[t[b].ls].sum - t[t[a].ls].sum;
    if (k < s) return kth(t[a].ls, t[b].ls, l, m, k);
    return kth(t[a].rs, t[b].rs, m, r, k - s);
}

int kth(int l, int r, int k) {
    assert(0 <= l && l <= r && r < SZ(rt));
    assert(0 <= k && k < t[rt[r]].sum - t[rt[l]].sum);
    return kth(rt[l], rt[r], 0, n, k);
}
};

```

```

vector<int> xs = a;
sort(ALL(xs));
xs.erase(unique(ALL(xs)), xs.end());
Chairman seg(SZ(xs));
for (int x : a) {
    seg.add(lower_bound(ALL(xs), x) - xs.begin());
}
int p = seg.kth(l, r, k);
int ans = xs[p];

```

3.5 Li Chao Tree

- 维护整数横坐标 $[L, R)$ 上直线的最小值，支持整条直线或限制在 $[l, r)$ 内的线段插入。
- add($\{k, b\}$) 插入 $y = kx + b$, qry(x) 查询横坐标 x 处的最小值；求最大值时将直线和答案同时取反。
- 动态开点，空间复杂度与实际插入产生的节点数有关。计算结果会被截断到 $[-4 \times 10^{18}, 4 \times 10^{18}]$ 。

```

struct LiChao {
    static constexpr LL INF = (LL)4e18;
    struct Line {
        LL k, b;
        Line(LL k = 0, LL b = INF) : k(k), b(b) {}
        LL get(LL x) {
            I y = (I)k * x + b;
            if (y > INF) return INF;
            if (y < -INF) return -INF;
            return y;
        }
    };
    struct Node {
        Line f;
        int ls = 0, rs = 0;
    };
    int L, R;
    vector<Node> t = {Node(), Node()};
    LiChao(int L, int R) : L(L), R(R) { assert(L < R); }
    int node() {
        t.emplace_back();
        return SZ(t) - 1;
    }
    void add(Line f, int p, int l, int r) {
        int m = l + (r - l) / 2;
        bool x = f.get(l) < t[p].f.get(l);
        bool y = f.get(m) < t[p].f.get(m);
        if (y) swap(f, t[p].f);
        if (r - l == 1) return;
        if (x != y) {
            if (!t[p].ls) t[p].ls = node();
            add(f, t[p].ls, l, m);
        } else {
            if (!t[p].rs) t[p].rs = node();
            add(f, t[p].rs, m, r);
        }
    }
    void add(Line f) { add(f, 1, L, R); }
    void add(int ql, int qr, Line f, int p, int l, int r) {
        if (qr <= l || r <= ql) return;
    }
}

```

```

    if (ql <= l && r <= qr) return add(f, p, l, r);
    int m = l + (r - l) / 2;
    if (ql < m) {
        if (!t[p].ls) t[p].ls = node();
        add(ql, qr, f, t[p].ls, l, m);
    }
    if (m < qr) {
        if (!t[p].rs) t[p].rs = node();
        add(ql, qr, f, t[p].rs, m, r);
    }
}

void add(int l, int r, Line f) {
    assert(L <= l && l <= r && r <= R);
    if (l < r) add(l, r, f, 1, L, R);
}

LL qry(int x, int p, int l, int r) {
    if (!p) return INF;
    LL ans = t[p].f.get(x);
    if (r - l == 1) return ans;
    int m = l + (r - l) / 2;
    if (x < m) return min(ans, qry(x, t[p].ls, l, m));
    return min(ans, qry(x, t[p].rs, m, r));
}

LL qry(int x) {
    assert(L <= x && x < R);
    return qry(x, 1, L, R);
}
};

```

3.6 Segment Tree Beats

- 所有位置和区间均为 0-index，区间使用半开形式 $[l, r)$ 。
- 支持区间加、区间 chmin、区间 chmax，以及区间和、最大值、最小值查询。
- chmin/chmax 利用严格次大值和严格次小值剪枝，复杂度为线段树 Beats 的均摊复杂度。

```

struct Beats {
    static constexpr LL INF = (LL)4e18;
    struct Node {
        LL sum = 0, add = 0;
        LL mx = -INF, smx = -INF;
        LL mn = INF, smn = INF;
        int cmx = 0, cmn = 0;
    };
    int n;
    vector<Node> t;
    Beats(const vector<LL>& a) : n(a.size()), t(4 * n) {
        assert(n > 0);
        build(1, 0, n, a);
    }
};

```

```

}

Node merge(Node a, Node b) {
    Node c;
    c.sum = a.sum + b.sum;
    if (a.mx == b.mx) {
        c.mx = a.mx;
        c.smx = max(a.smx, b.smx);
        c.cmx = a.cmx + b.cmx;
    } else if (a.mx > b.mx) {
        c.mx = a.mx;
        c.smx = max(a.smx, b.mx);
        c.cmx = a.cmx;
    } else {
        c.mx = b.mx;
        c.smx = max(a.mx, b.smx);
        c.cmx = b.cmx;
    }
    if (a.mn == b.mn) {
        c.mn = a.mn;
        c.smn = min(a.smn, b.smn);
        c.cmn = a.cmn + b.cmn;
    } else if (a.mn < b.mn) {
        c.mn = a.mn;
        c.smn = min(a.smn, b.mn);
        c.cmn = a.cmn;
    } else {
        c.mn = b.mn;
        c.smn = min(a.mn, b.smn);
        c.cmn = b.cmn;
    }
    return c;
}

void build(int p, int l, int r, const vector<LL>& a) {
    if (r - l == 1) {
        t[p].sum = t[p].mx = t[p].mn = a[l];
        t[p].cmx = t[p].cmn = 1;
        return;
    }
    int m = (l + r) / 2;
    build(2 * p, l, m, a);
    build(2 * p + 1, m, r, a);
    t[p] = merge(t[2 * p], t[2 * p + 1]);
}

void apply_add(int p, int len, LL x) {
    t[p].sum += x * len;
    t[p].mx += x;
    t[p].mn += x;
    if (t[p].smx != -INF) t[p].smx += x;
    if (t[p].smn != INF) t[p].smn += x;
    t[p].add += x;
}

```



```

}
void apply_chmin(int p, LL x) {
    if (t[p].mx <= x) return;
    t[p].sum += (x - t[p].mx) * t[p].cmx;
    if (t[p].mn == t[p].mx) t[p].mn = x;
    else if (t[p].smn == t[p].mx) t[p].smn = x;
    t[p].mx = x;
}
void apply_chmax(int p, LL x) {
    if (t[p].mn >= x) return;
    t[p].sum += (x - t[p].mn) * t[p].cmn;
    if (t[p].mx == t[p].mn) t[p].mx = x;
    else if (t[p].smx == t[p].mn) t[p].smx = x;
    t[p].mn = x;
}
void down(int p, int l, int r) {
    int m = (l + r) / 2;
    if (t[p].add) {
        apply_add(2 * p, m - l, t[p].add);
        apply_add(2 * p + 1, r - m, t[p].add);
        t[p].add = 0;
    }
    if (t[2 * p].mx > t[p].mx) apply_chmin(2 * p, t[p].mx);
    if (t[2 * p + 1].mx > t[p].mx) apply_chmin(2 * p + 1, t[p].mx);
    if (t[2 * p].mn < t[p].mn) apply_chmax(2 * p, t[p].mn);
    if (t[2 * p + 1].mn < t[p].mn) apply_chmax(2 * p + 1, t[p].mn);
}
void up(int p) {
    t[p] = merge(t[2 * p], t[2 * p + 1]);
}
void add(int ql, int qr, LL x, int p, int l, int r) {
    if (qr <= l || r <= ql) return;
    if (ql <= l && r <= qr) return apply_add(p, r - l, x);
    down(p, l, r);
    int m = (l + r) / 2;
    add(ql, qr, x, 2 * p, l, m);
    add(ql, qr, x, 2 * p + 1, m, r);
    up(p);
}
void chmin(int ql, int qr, LL x, int p, int l, int r) {
    if (qr <= l || r <= ql || t[p].mx <= x) return;
    if (ql <= l && r <= qr && t[p].smx < x) {
        return apply_chmin(p, x);
    }
    down(p, l, r);
    int m = (l + r) / 2;
    chmin(ql, qr, x, 2 * p, l, m);
    chmin(ql, qr, x, 2 * p + 1, m, r);
    up(p);
}

```

```

void chmax(int ql, int qr, LL x, int p, int l, int r) {
    if (qr <= l || r <= ql || t[p].mn >= x) return;
    if (ql <= l && r <= qr && t[p].smn > x) {
        return apply_chmax(p, x);
    }
    down(p, l, r);
    int m = (l + r) / 2;
    chmax(ql, qr, x, 2 * p, l, m);
    chmax(ql, qr, x, 2 * p + 1, m, r);
    up(p);
}
LL qsum(int ql, int qr, int p, int l, int r) {
    if (qr <= l || r <= ql) return 0;
    if (ql <= l && r <= qr) return t[p].sum;
    down(p, l, r);
    int m = (l + r) / 2;
    return qsum(ql, qr, 2 * p, l, m)
        + qsum(ql, qr, 2 * p + 1, m, r);
}
LL qmax(int ql, int qr, int p, int l, int r) {
    if (qr <= l || r <= ql) return -INF;
    if (ql <= l && r <= qr) return t[p].mx;
    down(p, l, r);
    int m = (l + r) / 2;
    return max(qmax(ql, qr, 2 * p, l, m),
        qmax(ql, qr, 2 * p + 1, m, r));
}
LL qmin(int ql, int qr, int p, int l, int r) {
    if (qr <= l || r <= ql) return INF;
    if (ql <= l && r <= qr) return t[p].mn;
    down(p, l, r);
    int m = (l + r) / 2;
    return min(qmin(ql, qr, 2 * p, l, m),
        qmin(ql, qr, 2 * p + 1, m, r));
}
void add(int l, int r, LL x) {
    assert(0 <= l && l <= r && r <= n);
    if (l < r) add(l, r, x, 1, 0, n);
}
void chmin(int l, int r, LL x) {
    assert(0 <= l && l <= r && r <= n);
    if (l < r) chmin(l, r, x, 1, 0, n);
}
void chmax(int l, int r, LL x) {
    assert(0 <= l && l <= r && r <= n);
    if (l < r) chmax(l, r, x, 1, 0, n);
}
LL qsum(int l, int r) {
    assert(0 <= l && l < r && r <= n);
    return qsum(l, r, 1, 0, n);
}

```

```

}
LL qmax(int l, int r) {
    assert(0 <= l && l < r && r <= n);
    return qmax(l, r, 1, 0, n);
}
LL qmin(int l, int r) {
    assert(0 <= l && l < r && r <= n);
    return qmin(l, r, 1, 0, n);
}
};

```

3.7 Segment Tree Merge / Split

- 动态开点维护值域 $[0, n)$ 上的非负计数；根节点用一个外部 `int` 保存，空树根为 `0`。
- `merge(a, b)` 将两棵树破坏性合并并返回新根；`split(p, l, r)` 从 `p` 中取走 $[l, r)$ 并返回取出的根。
- `kth(p, k)` 查询第 k 小值域下标， k 从 0 开始。若需支持该操作，应保证所有叶子计数非负。

```

struct SplitSGT {
    struct Node {
        int ls = 0, rs = 0;
        LL sum = 0;
    };
    int n;
    vector<Node> t = {Node()};
    SplitSGT(int n) : n(n) { assert(n > 0); }
    int node() {
        t.emplace_back();
        return SZ(t) - 1;
    }
    void up(int p) {
        t[p].sum = t[t[p].ls].sum + t[t[p].rs].sum;
    }
    int add(int p, int l, int r, int x, LL v) {
        if (!p) p = node();
        if (r - l == 1) {
            t[p].sum += v;
            return p;
        }
        int m = (l + r) / 2;
        if (x < m) t[p].ls = add(t[p].ls, l, m, x, v);
        else t[p].rs = add(t[p].rs, m, r, x, v);
        up(p);
        return p;
    }
    void add(int& p, int x, LL v) {
        assert(0 <= x && x < n);
        p = add(p, 0, n, x, v);
    }
};

```

```

}
LL sum(int p, int l, int r, int ql, int qr) {
    if (!p || qr <= l || r <= ql) return 0;
    if (ql <= l && r <= qr) return t[p].sum;
    int m = (l + r) / 2;
    return sum(t[p].ls, l, m, ql, qr)
        + sum(t[p].rs, m, r, ql, qr);
}
LL sum(int p, int l, int r) {
    assert(0 <= l && l <= r && r <= n);
    return sum(p, 0, n, l, r);
}
int merge(int a, int b, int l, int r) {
    if (!a || !b) return a + b;
    if (r - l == 1) {
        t[a].sum += t[b].sum;
        return a;
    }
    int m = (l + r) / 2;
    t[a].ls = merge(t[a].ls, t[b].ls, l, m);
    t[a].rs = merge(t[a].rs, t[b].rs, m, r);
    up(a);
    return a;
}
int merge(int a, int b) { return merge(a, b, 0, n); }
pair<int, int> split(int p, int l, int r, int ql, int qr) {
    if (!p || qr <= l || r <= ql) return {p, 0};
    if (ql <= l && r <= qr) return {0, p};
    int m = (l + r) / 2;
    auto [al, bl] = split(t[p].ls, l, m, ql, qr);
    auto [ar, br] = split(t[p].rs, m, r, ql, qr);
    t[p].ls = al;
    t[p].rs = ar;
    up(p);
    if (!al && !ar) p = 0;
    if (!bl && !br) return {p, 0};
    int q = node();
    t[q].ls = bl;
    t[q].rs = br;
    up(q);
    return {p, q};
}
int split(int& p, int l, int r) {
    assert(0 <= l && l <= r && r <= n);
    auto [a, b] = split(p, 0, n, l, r);
    p = a;
    return b;
}
int kth(int p, LL k) {
    assert(p && 0 <= k && k < t[p].sum);
}

```

```

int l = 0, r = n;
while (r - l > 1) {
    int m = (l + r) / 2;
    LL s = t[t[p].ls].sum;
    if (k < s) {
        p = t[p].ls;
        r = m;
    } else {
        k -= s;
        p = t[p].rs;
        l = m;
    }
}
return l;
};

```

4 String

4.1 KMP

- $kmp(s)[i]$ 表示 $s[0, i)$ 的最长相等真前后缀长度，返回数组长度为 $n+1$ 。
- 模板只依赖 `size()` 和元素判等，可直接用于字符串或一般序列。

```

template <class T> vector<int> kmp(const T& s) {
    int n = s.size();
    vector<int> f(n + 1);
    for (int i = 1, j = 0; i < n; i++) {
        while (j && s[i] != s[j]) j = f[j];
        if (s[i] == s[j]) j++;
        f[i + 1] = j;
    }
    return f;
}

```

4.2 Z Function

- $ZFunc(s)[i]$ 表示 s 与后缀 $s[i, n)$ 的最长公共前缀长度，并令 $z[0]=n$ 。
- $exKMP(s, t, z)$ 返回每个后缀 $s[i, n)$ 与模式串 t 的最长公共前缀长度，其中 $z=ZFunc(s) \hookrightarrow t$ 。

```

template <class T> vector<int> ZFunc(const T& s) {
    int n = s.size(), l = 0, r = 0;
    vector<int> z(n);
    if (!n) return z;
    z[0] = n;
    for (int i = 1; i < n; i++) {
        if (i <= r && z[i - l] < r - i + 1) z[i] = z[i - l];
        else {

```

```

            z[i] = max(0, r - i + 1);
            while (i + z[i] < n && s[i + z[i]] == s[z[i]]) z[i]++;
            if (i + z[i] - 1 > r) l = i, r = i + z[i] - 1;
        }
    }
    return z;
}

template <class T>
vector<int> exKMP(const T& s, const T& t, const vector<int>& z) {
    int n = s.size(), m = t.size(), l = 0, r = 0;
    vector<int> p(n);
    if (!n) return p;
    while (p[0] < n && p[0] < m && s[p[0]] == t[p[0]]) p[0]++;
    for (int i = 1; i < n; i++) {
        if (i <= r && z[i - l] < r - i + 1) p[i] = z[i - l];
        else {
            p[i] = max(0, r - i + 1);
            while (i + p[i] < n && p[i] < m && s[i + p[i]] == t[p[i]]) p[i]++;
            if (i + p[i] - 1 > r) l = i, r = i + p[i] - 1;
        }
    }
    return p;
}

```

4.3 String Hash

- 双模 Hash，所有位置均为 0-index， $get(l, r)$ 返回半开区间 $s[l, r)$ 的归一化 Hash。
- 两段字符串长度相同且 Hash 相等时可认为相等；Hash 算法始终存在极小碰撞概率。

```

struct StringHash {
    using H = pair<LL, LL>;
    static constexpr LL P1 = 2034452107, P2 = 2013074419;
    static constexpr LL B1 = 137, B2 = 149;
    int n;
    vector<LL> pw1, pw2, h1, h2;
    StringHash(const string& s) : n(s.size()),
        pw1(n + 1, 1), pw2(n + 1, 1), h1(n + 1), h2(n + 1) {
        for (int i = 0; i < n; i++) {
            int x = (unsigned char)s[i] + 1;
            pw1[i + 1] = pw1[i] * B1 % P1;
            pw2[i + 1] = pw2[i] * B2 % P2;
            h1[i + 1] = (h1[i] * B1 + x) % P1;
            h2[i + 1] = (h2[i] * B2 + x) % P2;
        }
    }
    H get(int l, int r) {
        assert(0 <= l && l <= r && r <= n);
        LL x = (h1[r] - h1[l] * pw1[r - l] % P1 + P1) % P1;

```

```

    LL y = (h2[r] - h2[l] * pw2[r - l] % P2 + P2) % P2;
    return {x, y};
}
};

```

4.4 Manacher

- 算法在变换串 $\#s[0]\#s[1]\#\dots\#$ 上运行, $p[i]-1$ 是以变换串位置 i 为中心的最长回文子串在原串中的长度。
- 对应原串半开区间为 $[(i-p[i]+1)/2, (i+p[i]-1)/2)$ 。

```

vector<int> Manacher(const string& s) {
    string t = "#";
    for (char c : s) t += c, t += '#';
    int n = t.size();
    vector<int> p(n);
    for (int i = 0, l = 0, r = 0; i < n; i++) {
        p[i] = i < r ? min(p[2 * l - i], r - i) : 0;
        while (i + p[i] < n && i - p[i] >= 0
            && t[i + p[i]] == t[i - p[i]]) p[i]++;
        if (i + p[i] > r) l = i, r = i + p[i];
    }
    return p;
}

```

4.5 Aho-Corasick

- 字符集固定为小写字母; 先用 `add(s)` 插入全部模式串并保存返回的终止节点, 再调用一次 `build()`。
- `count(s)[p]` 是节点 p 对应模式串在文本 s 中的出现次数, 重复模式串会共享终止节点。

```

struct ACAM {
    struct Node {
        array<int, 26> ch{};
        int fail = 0;
    };
    vector<Node> t = {};
    vector<int> ord;
    ACAM(int n = 0) { t.reserve(n + 1); }
    int add(const string& s) {
        int p = 0;
        for (char x : s) {
            int c = x - 'a';
            assert(0 <= c && c < 26);
            if (!t[p].ch[c]) {
                t[p].ch[c] = t.size();
                t.push_back({});
            }
        }
    }
}

```

```

    p = t[p].ch[c];
}
return p;
}

void build() {
    queue<int> q;
    ord.clear();
    for (int c = 0; c < 26; c++) {
        if (t[0].ch[c]) q.push(t[0].ch[c]);
    }
    while (SZ(q)) {
        int u = q.front();
        q.pop();
        ord.pb(u);
        for (int c = 0; c < 26; c++) {
            int v = t[u].ch[c];
            if (v) {
                t[v].fail = t[t[u].fail].ch[c];
                q.push(v);
            } else {
                t[u].ch[c] = t[t[u].fail].ch[c];
            }
        }
    }
}

vector<LL> count(const string& s) {
    vector<LL> cnt(t.size());
    int p = 0;
    for (char x : s) {
        int c = x - 'a';
        assert(0 <= c && c < 26);
        p = t[p].ch[c];
        cnt[p]++;
    }
    for (int i = SZ(ord) - 1; i >= 0; i--) {
        int u = ord[i];
        cnt[t[u].fail] += cnt[u];
    }
    return cnt;
}
};

```

4.6 Suffix Array

- $sa[i]$ 是排名第 i 的后缀起点, $rk[i]$ 是后缀 $s[i, n)$ 的排名, 均为 0-index。
- $lc[i]$ 等于 $\text{lcp}(s[sa[i], n), s[sa[i+1], n))$, 数组长度为 $n-1$ 。

```

struct SuffixArray {
    int n;
}

```

```

vector<int> sa, rk, lc;
SuffixArray(const string& s) : n(s.size()), sa(n), rk(n), lc(max(0, n - 1)) {
    if (!n) return;
    iota(ALL(sa), 0);
    sort(ALL(sa), [&](int a, int b) { return s[a] < s[b]; });
    rk[sa[0]] = 0;
    for (int i = 1; i < n; i++) {
        rk[sa[i]] = rk[sa[i - 1]] + (s[sa[i]] != s[sa[i - 1]]);
    }
    vector<int> tmp, cnt(n);
    tmp.reserve(n);
    for (int w = 1; rk[sa[n - 1]] < n - 1; w *= 2) {
        tmp.clear();
        for (int i = max(0, n - w); i < n; i++) tmp.pb(i);
        for (int i = 0; i < n; i++) if (sa[i] >= w) tmp.pb(sa[i] - w);
        fill(ALL(cnt), 0);
        for (int i = 0; i < n; i++) cnt[rk[i]]++;
        partial_sum(ALL(cnt), cnt.begin());
        for (int i = n - 1; i >= 0; i--) sa[--cnt[rk[tmp[i]]]] = tmp[i];
        swap(rk, tmp);
        rk[sa[0]] = 0;
        for (int i = 1; i < n; i++) {
            int a = sa[i - 1], b = sa[i];
            int x = a + w < n ? tmp[a + w] : -1;
            int y = b + w < n ? tmp[b + w] : -1;
            rk[b] = rk[a] + (tmp[a] != tmp[b] || x != y);
        }
    }
    for (int i = 0, k = 0; i < n; i++) {
        int j = rk[i];
        if (!j) {
            k = 0;
            continue;
        }
        if (k) k--;
        int p = sa[j - 1];
        while (i + k < n && p + k < n && s[i + k] == s[p + k]) k++;
        lc[j - 1] = k;
    }
};

```

4.7 Suffix Automaton

- 根节点编号为 1；普通字符串从 $p=1$ 开始，依次令 $p=\text{sam.ins}(p, c-\text{'a'})$ 。
- 最多产生 $2n$ 个节点；fa 是后缀链接，len 是状态所表示子串的最大长度。

```

template <int C = 26>
struct SAM {

```

```

    vector<array<int, C>> ch;
    vector<int> fa, len;
    SAM(int n = 0) : ch(2), fa(2), len(2) {
        ch.reserve(2 * n + 2);
        fa.reserve(2 * n + 2);
        len.reserve(2 * n + 2);
    }
    int node() {
        ch.push_back({});
        fa.pb(0);
        len.pb(0);
        return SZ(ch) - 1;
    }
    int ins(int p, int c) {
        assert(0 <= c && c < C);
        int cur = node();
        len[cur] = len[p] + 1;
        for (; p && !ch[p][c]; p = fa[p]) ch[p][c] = cur;
        if (!p) fa[cur] = 1;
        else {
            int q = ch[p][c];
            if (len[q] == len[p] + 1) fa[cur] = q;
            else {
                int cl = node();
                len[cl] = len[p] + 1;
                ch[cl] = ch[q];
                fa[cl] = fa[q];
                fa[q] = fa[cur] = cl;
                for (; p && ch[p][c] == q; p = fa[p]) ch[p][c] = cl;
            }
        }
        return cur;
    }
};

```

4.8 Palindromic Tree

- 字符须先映射到 $[0, C)$ ；调用 $\text{init}(n)$ 后逐个 $\text{add}(c)$ ，节点 0、1 分别是长度 0、-1 的根。
- $\text{size}()$ 是本质不同回文串数， $\text{suf_cnt}()$ 是当前回文后缀数， $\text{count}()$ 是已经插入串的回文子串总数。
- 全部插入后调用一次 $\text{calc}()$ ，即可由 $t[p].\text{cnt}$ 得到每个回文串的出现次数。

```

template <int C = 26>
struct PAM {
    struct Node {
        array<int, C> ch{};
        int fail = 0, len = 0, cnt = 0, num = 0;
    };

```

```

vector<Node> t;
vector<int> s;
int las;
LL tot;
void init(int n) {
    t.assign(2, {});
    t.reserve(n + 2);
    t[0].fail = t[1].fail = 1;
    t[1].len = -1;
    s.assign(1, -1);
    s.reserve(n + 1);
    las = 0;
    tot = 0;
}
int get_fail(int p, int c) {
    int n = SZ(s) - 1;
    while (s[n - t[p].len - 1] != c) p = t[p].fail;
    return p;
}
int add(int c) {
    assert(0 <= c && c < C);
    s.pb(c);
    int p = get_fail(las, c);
    if (!t[p].ch[c]) {
        int q = get_fail(t[p].fail, c);
        Node x;
        x.len = t[p].len + 2;
        x.fail = t[q].ch[c];
        x.num = t[x.fail].num + 1;
        t[p].ch[c] = SZ(t);
        t.pb(x);
    }
    las = t[p].ch[c];
    t[las].cnt++;
    tot += t[las].num;
    return las;
}
int size() { return SZ(t) - 2; }
int suf_cnt() { return t[las].num; }
LL count() { return tot; }
void calc() {
    for (int p = SZ(t) - 1; p >= 2; p--) t[t[p].fail].cnt += t[p].cnt;
}
};

```

4.9 Lyndon / Minimum Representation

- Duval(s) 返回 Lyndon 分解的 0-index 半开区间，按顺序拼接后恰为原序列。
- min_rep(s) 返回循环同构序列中字典序最小表示的起点；空序列返回 0。

```

template <class T>
vector<pair<int, int>> Duval(const T& s) {
    int n = s.size(), i = 0;
    vector<pair<int, int>> a;
    while (i < n) {
        int j = i, k = i + 1;
        while (k < n && s[j] <= s[k]) {
            if (s[j] < s[k]) j = i;
            else j++;
            k++;
        }
        while (i <= j) {
            int r = i + k - j;
            a.push_back({i, r});
            i = r;
        }
    }
    return a;
}

template <class T> int min_rep(const T& s) {
    int n = s.size(), i = 0, j = 1, k = 0;
    if (!n) return 0;
    while (i < n && j < n && k < n) {
        if (s[(i + k) % n] == s[(j + k) % n]) k++;
        else {
            if (s[(i + k) % n] > s[(j + k) % n]) i += k + 1;
            else j += k + 1;
            if (i == j) j++;
            k = 0;
        }
    }
    return min(i, j);
}

```

5 Geometry

5.1 vector

```

using db = long double;
struct p2 {
    db x, y;
    db len() { return hypot(x, y); }
    p2 operator+(p2 b) { return {x + b.x, y + b.y}; }
    p2 operator-(p2 b) { return {x - b.x, y - b.y}; }
    p2 operator*(db k) { return {x * k, y * k}; }
    p2 operator/(db k) { return {x / k, y / k}; }
    db operator*(p2 b) { return x * b.x + y * b.y; }
    db operator%(p2 b) { return x * b.y - y * b.x; }
}

```

```
};
using ps = vector<p2>;
```

5.2 seg_poly

```
// pa 到 pb 的叉积，逆时针转为正
db side(p2 p, p2 a, p2 b) { return (a - p) % (b - p); }
db toL(p2 p, p2 a, p2 b) { return abs(side(p, a, b)) / (b - a).len(); }
```

6 Number Theory

6.1 ModInt

- 修改 P 可更换模数；除法和 `inv()` 要求 P 为质数且除数非零。
- `qp(a,b)` 要求指数 $b \geq 0$ ，也可用于其他定义了乘法的类型。

```
constexpr int P = 998244353;
template <class T> T qp(T a, LL b) {
    assert(b >= 0);
    T r{1};
    for (; b >>= 1, a *= a if (b & 1) r *= a;
        return r;
}
struct Mint {
    int v;
    Mint(LL x = 0) : v(int((x % P + P) % P)) {}
    int val() const { return v; }
    Mint operator-(const Mint& b) { return Mint(-v); }
    Mint& operator+=(const Mint& b) {
        if ((v += b.v) >= P) v -= P;
        return *this;
    }
    Mint& operator-=(const Mint& b) {
        if ((v -= b.v) < 0) v += P;
        return *this;
    }
    Mint& operator*=(const Mint& b) {
        v = int(LL(v) * b.v % P);
        return *this;
    }
    Mint& operator/=(const Mint& b) { return *this *= b.inv(); }
    friend Mint operator+(Mint a, const Mint& b) { return a += b; }
    friend Mint operator-(Mint a, const Mint& b) { return a -= b; }
    friend Mint operator*(Mint a, const Mint& b) { return a *= b; }
    friend Mint operator/(Mint a, const Mint& b) { return a /= b; }
    Mint inv() const {
        assert(v);
        return qp(*this, P - 2);
    }
};
```

```
}
friend istream& operator>>(istream& is, Mint& a) {
    LL x;
    is >> x;
    return a = Mint(x), is;
}
friend ostream& operator<<(ostream& os, const Mint& a) {
    return os << a.v;
}
};
using mint = Mint;
```

6.2 Extended GCD

- `exgcd(a,b,x,y)` 要求 $a, b \geq 0$ 且不同时为零，返回 $g = \gcd(a, b)$ ，并满足 $ax + by = g$ 。
- `linear_congruence(a,b,m)` 求 $ax \equiv b \pmod{m}$ 。无解返回 `nullopt`；否则返回 $\{r, p\}$ ，全部解为 $x \equiv r \pmod{p}$ 。

```
LL exgcd(LL a, LL b, LL& x, LL& y) {
    assert(a >= 0 && b >= 0 && (a || b));
    if (!b) return x = 1, y = 0, a;
    LL g = exgcd(b, a % b, y, x);
    y -= a / b * x;
    return g;
}
optional<pair<LL, LL>> linear_congruence(LL a, LL b, LL m) {
    assert(m > 0);
    a %= m;
    b %= m;
    if (a < 0) a += m;
    if (b < 0) b += m;
    LL x, y, g = exgcd(a, m, x, y);
    if (b % g) return nullopt;
    LL p = m / g;
    LL r = I(x) * (b / g) % p;
    if (r < 0) r += p;
    return pair<LL, LL>{LL(r), p};
}
```

6.3 CRT

- 合并任意个方程 $x \equiv r_i \pmod{m_i}$ ，模数不要求互质，但必须为正数。
- 返回 $\{r, m\}$ 表示全部解为 $x \equiv r \pmod{m}$ ；无解返回 $\{0, 0\}$ 。要求最终模数最小公倍数不超过 `LL`。

```
LL safe_mod(LL x, LL m) {
    x %= m;
    return x < 0 ? x + m : x;
}
```

```

pair<LL, LL> inv_gcd(LL a, LL b) {
    a = safe_mod(a, b);
    if (!a) return {b, 0};
    LL s = b, t = a;
    I x = 0, y = 1;
    while (t) {
        LL q = s / t;
        s -= q * t;
        swap(s, t);
        x -= I(q) * y;
        swap(x, y);
    }
    LL m = b / s;
    x %= m;
    if (x < 0) x += m;
    return {s, LL(x)};
}

pair<LL, LL> crt(const vector<LL>& r, const vector<LL>& m) {
    assert(r.size() == m.size());
    LL r0 = 0, m0 = 1;
    for (int i = 0; i < SZ(r); i++) {
        assert(m[i] > 0);
        LL r1 = safe_mod(r[i], m[i]), m1 = m[i];
        if (m0 < m1) swap(r0, r1), swap(m0, m1);
        if (m0 % m1 == 0) {
            if (r0 % m1 != r1) return {0, 0};
            continue;
        }
        auto [g, im] = inv_gcd(m0, m1);
        LL u = m1 / g;
        if ((r1 - r0) % g) return {0, 0};
        LL x = LL(I((r1 - r0) / g) * im % u);
        I nm = I(m0) * u;
        assert(nm <= numeric_limits<LL>::max());
        I nr = r0 + I(x) * m0;
        m0 = LL(nm);
        r0 = LL(nr % m0);
        if (r0 < 0) r0 += m0;
    }
    return {r0, m0};
}

```

6.4 Linear Sieve

- $O(n)$ 求出不超过 n 的全部质数、最小质因子、欧拉函数 ϕ 和莫比乌斯函数 μ 。
- $\text{factor}(x)$ 适用于 $1 \leq x \leq n$, 返回 {质因子, 次数}。

```

struct Sieve {
    int n;

```

```

vector<int> pr, minp, phi, mu;
Sieve(int n_) : n(n_), minp(n + 1), phi(n + 1), mu(n + 1) {
    assert(n >= 1);
    phi[1] = mu[1] = 1;
    for (int i = 2; i <= n; i++) {
        if (!minp[i]) {
            pr.push_back(i);
            minp[i] = i;
            phi[i] = i - 1;
            mu[i] = -1;
        }
        for (int p : pr) {
            if (LL(i) * p > n) break;
            minp[i * p] = p;
            if (i % p == 0) {
                phi[i * p] = phi[i] * p;
                break;
            }
            phi[i * p] = phi[i] * (p - 1);
            mu[i * p] = -mu[i];
        }
    }
}

vector<pair<int, int>> factor(int x) const {
    assert(1 <= x && x <= n);
    vector<pair<int, int>> f;
    while (x > 1) {
        int p = minp[x], c = 0;
        do x /= p, c++;
        while (x > 1 && minp[x] == p);
        f.push_back({p, c});
    }
    return f;
}
};

```

6.5 Miller-Rabin / Pollard-Rho

- $\text{prime}(n)$ 对整个 ULL 范围使用确定性 Miller-Rabin; $\text{factor}(n)$ 返回带重数且升序排列的质因子。
- 乘法使用 `__uint128_t`, 要求编译器支持 GNU C++; $\text{factor}(1)$ 返回空数组。

```

namespace Factor {
using U128 = __uint128_t;
ULL mul_mod(ULL a, ULL b, ULL m) {
    return ULL(U128(a) * b % m);
}
ULL add_mod(ULL a, ULL b, ULL m) {
    return ULL((U128(a) + b) % m);
}

```



```

}
ULL pow_mod(ULL a, ULL b, ULL m) {
    ULL r = 1 % m;
    for (; b >>= 1, a = mul_mod(a, a, m))
        if (b & 1) r = mul_mod(r, a, m);
    return r;
}
bool prime(ULL n) {
    if (n < 2) return false;
    for (ULL p : {2ULL, 3ULL, 5ULL, 7ULL, 11ULL, 13ULL, 17ULL,
        19ULL, 23ULL, 29ULL, 31ULL, 37ULL}) {
        if (n % p == 0) return n == p;
    }
    int s = __builtin_ctzll(n - 1);
    ULL d = (n - 1) >> s;
    for (ULL a : {2ULL, 325ULL, 9375ULL, 28178ULL,
        450775ULL, 9780504ULL, 1795265022ULL}) {
        if (a % n == 0) continue;
        ULL x = pow_mod(a % n, d, n);
        if (x == 1 || x == n - 1) continue;
        bool ok = false;
        for (int r = 1; r < s; r++) {
            x = mul_mod(x, x, n);
            if (x == n - 1) {
                ok = true;
                break;
            }
        }
        if (!ok) return false;
    }
    return true;
}
ULL diff(ULL a, ULL b) { return a > b ? a - b : b - a; }
ULL rho(ULL n) {
    if (n % 2 == 0) return 2;
    if (n % 3 == 0) return 3;
    static mt19937_64 rng(chrono::steady_clock::now()
        .time_since_epoch().count());
    while (true) {
        ULL y = rng() % (n - 2) + 2;
        ULL c = rng() % (n - 1) + 1;
        auto f = [&](ULL x) { return add_mod(mul_mod(x, x, n), c, n); };
        ULL g = 1, x = 0, z = 0;
        for (ULL r = 1; g == 1; r <= 1) {
            x = y;
            for (ULL i = 0; i < r; i++) y = f(y);
            for (ULL k = 0; k < r && g == 1; k += 128) {
                z = y;
                ULL q = 1;
                for (ULL i = 0; i < min<ULL>(128, r - k); i++) {

```

```

                    y = f(y);
                    q = mul_mod(q, diff(x, y), n);
                }
                g = gcd(q, n);
            }
        }
        if (g == n) {
            do z = f(z), g = gcd(diff(x, z), n);
            while (g == 1);
        }
        if (g != n) return g;
    }
}
void factor(ULL n, vector<ULL>& f) {
    if (n == 1) return;
    if (prime(n)) return f.push_back(n);
    ULL d = rho(n);
    factor(d, f);
    factor(n / d, f);
}
vector<ULL> factor(ULL n) {
    assert(n > 0);
    vector<ULL> f;
    factor(n, f);
    sort(f.begin(), f.end());
    return f;
}
vector<pair<ULL, int>> factor_count(ULL n) {
    vector<ULL> a = factor(n);
    vector<pair<ULL, int>> f;
    for (ULL p : a) {
        if (f.empty() || f.back().first != p) f.push_back({p, 1});
        else f.back().second++;
    }
    return f;
}
}
}

```

6.6 Primitive Root

- 求质数 p 的最小正原根，依赖前一个 Miller–Rabin / Pollard–Rho 条目。
- 这里只处理质数模数；`primitive_root(2)=1`。

```

ULL primitive_root(ULL p) {
    assert(Factor::prime(p));
    if (p == 2) return 1;
    vector<ULL> f = Factor::factor(p - 1);
    f.erase(unique(f.begin(), f.end()), f.end());
    for (ULL g = 2; g < p; g++) {

```

```

bool ok = true;
for (ULL q : f) {
    if (Factor::pow_mod(g, (p - 1) / q, p) == 1) {
        ok = false;
        break;
    }
}
if (ok) return g;
}
return 0;
}

```

6.7 exBSGS

- 求最小的 $x \geq 0$ 使 $a^x \equiv b \pmod{m}$ ，模数不要求为质数；无解返回-1。
- 时间与空间复杂度均为 $O(\sqrt{m})$ ，使用哈希表保存 baby steps。

```

namespace DiscreteLog {
LL norm(LL x, LL m) {
    x %= m;
    return x < 0 ? x + m : x;
}
LL mul_mod(LL a, LL b, LL m) { return LL(I(a) * b % m); }
LL inv_mod(LL a, LL m) {
    LL b = m;
    I x = 1, y = 0;
    while (b) {
        LL q = a / b;
        a -= q * b;
        swap(a, b);
        x -= I(q) * y;
        swap(x, y);
    }
    assert(a == 1);
    x %= m;
    if (x < 0) x += m;
    return LL(x);
}
LL bsgs(LL a, LL b, LL m) {
    assert(m > 1 && gcd(a, m) == 1);
    LL n = LL(sqrtl((long double)m)) + 1;
    unordered_map<LL, LL> baby;
    baby.reserve(size_t(2 * n + 1));
    LL cur = 1 % m;
    for (LL j = 0; j < n; j++) {
        baby.emplace(cur, j);
        cur = mul_mod(cur, a, m);
    }
    LL step = inv_mod(cur, m);

```

```

cur = b;
for (LL i = 0; i <= n; i++) {
    auto it = baby.find(cur);
    if (it != baby.end()) {
        I x = I(i) * n + it->second;
        if (x <= numeric_limits<LL>::max()) return LL(x);
        return -1;
    }
    cur = mul_mod(cur, step, m);
}
return -1;
}
LL exbsgs(LL a, LL b, LL m) {
    assert(m > 0);
    if (m == 1) return 0;
    a = norm(a, m);
    b = norm(b, m);
    if (b == 1) return 0;
    LL k = 1, add = 0;
    while (true) {
        LL g = gcd(a, m);
        if (g == 1) break;
        if (b % g) return -1;
        b /= g;
        m /= g;
        k = mul_mod(k, a / g, m);
        add++;
        if (k == b) return add;
    }
    b = mul_mod(b, inv_mod(k, m), m);
    LL x = bsgs(a, b, m);
    return x < 0 ? -1 : x + add;
}
}
using DiscreteLog::exbsgs;

```

6.8 Tonelli-Shanks

- $\text{mod_sqrt}(a, p)$ 要求 p 为质数，返回较小的平方根；二次非剩余返回-1。
- 另一根为 $p - r$ 。复杂度为 $O(\log^2 p)$ ，乘法使用 `__int128` 防止 LL 溢出。

```

LL sqrt_mul(LL a, LL b, LL p) { return LL(I(a) * b % p); }
LL sqrt_pow(LL a, LL b, LL p) {
    LL r = 1 % p;
    for (; b >>= 1, a = sqrt_mul(a, a, p))
        if (b & 1) r = sqrt_mul(r, a, p);
    return r;
}
LL mod_sqrt(LL a, LL p) {

```

```
assert(p >= 2);
a %= p;
if (a < 0) a += p;
if (!a || p == 2) return a;
if (sqrt_pow(a, (p - 1) / 2, p) != 1) return -1;
if (p % 4 == 3) {
    LL x = sqrt_pow(a, (p + 1) / 4, p);
    return min(x, p - x);
}
LL q = p - 1; int s = 0;
while (!(q & 1)) q >>= 1, s++;
LL z = 2; while (sqrt_pow(z, (p - 1) / 2, p) != p - 1) z++;
LL c = sqrt_pow(z, q, p), x = sqrt_pow(a, (q + 1) / 2, p);
LL t = sqrt_pow(a, q, p); int m = s;
while (t != 1) {
    int i = 0; LL y = t;
    while (y != 1 && i < m) y = sqrt_mul(y, y, p), i++;
    assert(i < m);
    LL b = c;
    for (int j = 0; j < m - i - 1; j++) b = sqrt_mul(b, b, p);
    x = sqrt_mul(x, b, p);
    c = sqrt_mul(b, b, p);
    t = sqrt_mul(t, c, p);
    m = i;
}
return min(x, p - x);
}
```